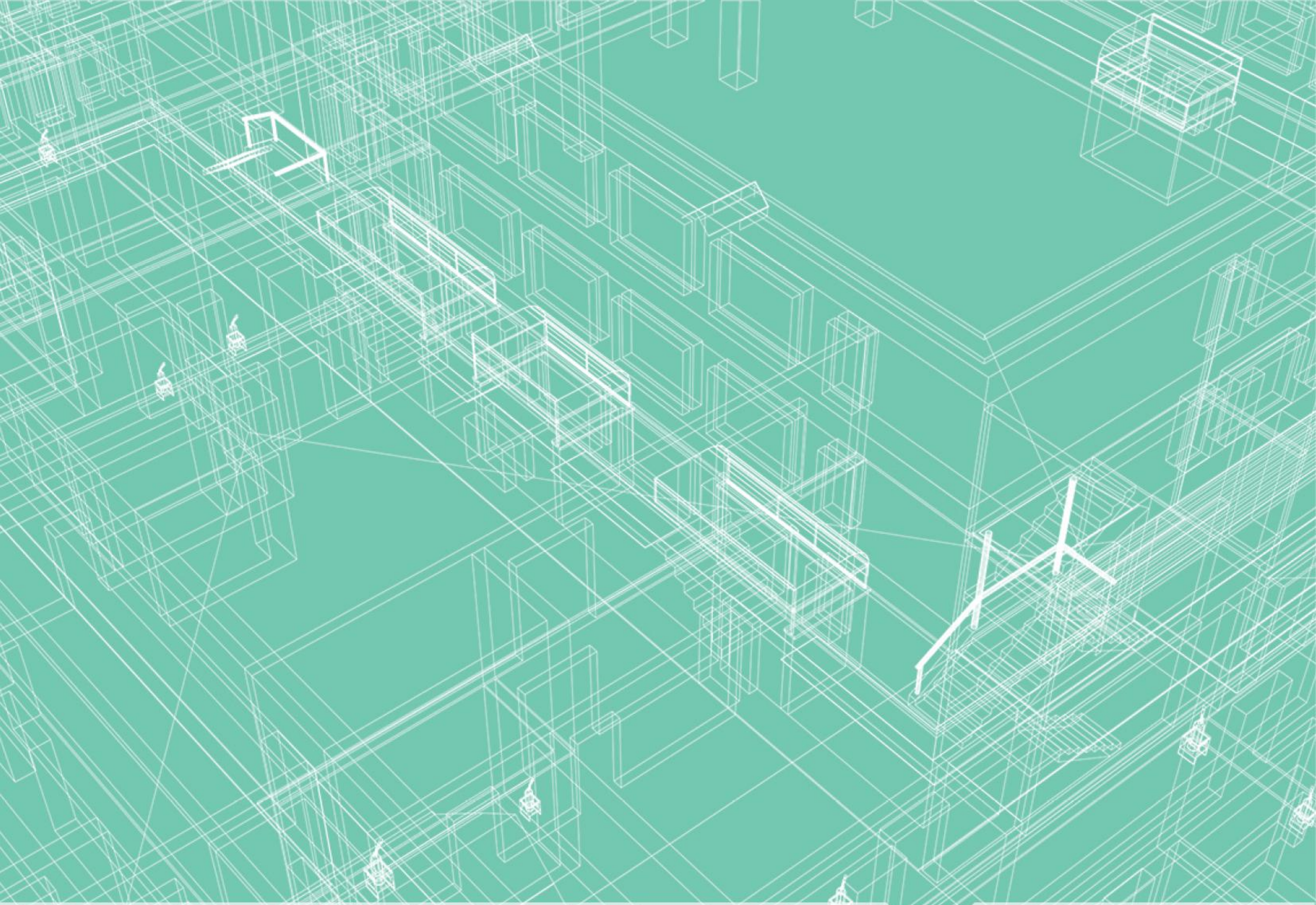

SECTION J – J1V3 REPORT

STANWELL OFFICE

ESD SERVICES

JHA

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DOCUMENT CONTROL SHEET

Project Number	Q250128
Project Name	Stanwell Office
Description	Section J – J1V3 Report
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Prepared By

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Revision History

Issue	Revision	Date
J1V3 Report	A	10/12/2025

CONTENTS

PURPOSE OF REPORT INCLUDING LIMITATIONS

This report identifies the results of engineering calculations undertaken by JHA to identify a pathway to building fabric energy efficiency compliance under NCC 2022 Section J.

NCC Section J intends that *'reduce energy consumption and energy peak demand, reduce greenhouse gas emissions and improve occupant health and amenity'* and applies to building fabric (J4), building sealing (J5) and building services (J6/J7, J8, J9).

It remains the responsibility of others to integrate energy efficiency considerations into the final building fabric design. For avoidance of doubt this report does not consider or provide any guarantee of:

- Thermal comfort in operation – refer architect, mechanical engineer and/or commission a detailed thermal comfort assessment.
- Acoustic comfort in operation – refer acoustic specialist.
- Building sealing and/or vapour barrier/permeability to mitigate condensation and mould risk - refer architect, facade and/or mechanical engineer.
- Fire performance and fire/smoke properties of construction materials –refer fire engineer, certifier and/or architect.
- Energy consumption in operation – NCC compliance provides no indication of future energy consumption.
- Durability and longevity of systems – refer structural and/or architect.
- Wind, glare, daylight and internal light quality – refer to relevant specialists.
- Renewable energy generation – refer to electrical engineer and/or specialist.
- Practicality or safety in construction/operation, budget compliance, aesthetic acceptability.

This report must not be misconstrued as architectural or façade design.

1. INTRODUCTION

This Section J – J1V3 report has been prepared to support compliance for building fabric for the Stanwell Office project.

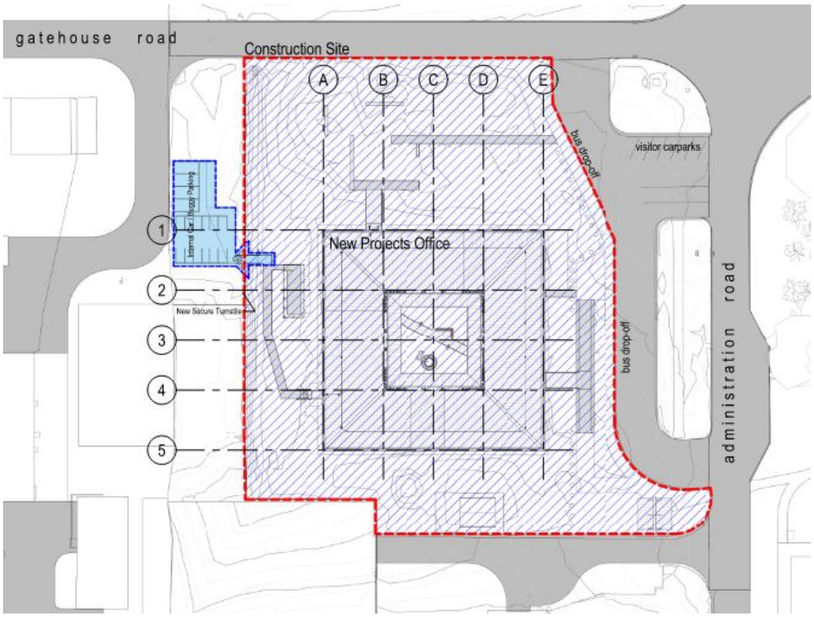

Figure 1 – Project Site

1.1 BASIS OF ASSESSMENT

The basis of assessment is tabulated below, we understand the following to be key objectives of this assessment:

- Identify minimum construction commitments to achieve Section J building fabric energy efficiency compliance for Part J4D4 (Roof & Ceiling), J4D6 (Walls & Glazing) and J4D7 (Floors).
- Reduce the floor insulation requirement.
- Reduce the glazing requirement.

Table 1 – Development Overview & Basis of Assessment

Property Name	Stanwell Office
Address	Switchyard Road, Stanwell, QLD, 4702
NCC Climate Zone	NCC climate zone 2
NCC Building Classification, Use & Compliance Pathway	Class 5 office (J1V3)
NCC & QDC Version	NCC 2022 Volume One
Compliance Elements Addressed under this Report	Energy Consumption of Proposed building is demonstrated via J1V3 modelling to be better than a DtS Reference building
Documentation Basis	See Appendix E

1.2 KEY RISKS

Please review the report and confirm any issue with the proposed construction commitment pathway.

2. PROJECT COMMITMENTS

2.1 BUILDING COMMITMENTS – MINIMUM PERFORMANCE REQUIREMENTS

Minimum building fabric thermal performance requirements are tabulated below. Please review carefully to confirm JHA's understanding of the building/construction is acceptable and will be included in the design documents and/or will be implemented in construction. This table must be read in conjunction with thermal envelope markups in Appendix A.

Table 2– Building Commitments – Minimum Performance Requirements

Building Element	Minimum Performance Requirements
Envelope Walls[1]	Total R-value: R1.9 Maximum Solar Absorptance: 0.5
External Glazing[2]	Total U-value: 4.2 SHGC[3]:0.64
Central Courtyard Glazing[2]	Total U-value: 4.7 SHGC[3]:0.59
Roof & Ceiling[1]	Total R-Value: R5.9 for a downward direction of heat flow. Maximum Solar Absorptance: 0.45
Suspended floor[1]	Total R-Value: R1.5 for a downward and upward direction of heat flow.
Slab on ground[4]	N/A

1. The specified thermal performance of the building fabric must be calculated including allowance for thermal bridging in accordance with AS/NZS 4859.2 as per NCC 2022 Part J4D3 or, Specification 37, or Specification 38 requirements. Interested parties are advised to confirm these values.
2. The specified thermal performance of the glazing includes the whole of window properties such as glass, glazed louvres and framing. The total thermal performance of the glazing. is anticipated to be officially confirmed by interested parties.
3. Lower SHGC can be achieved in many ways including with heavily tinted and/or dark glass. Glazing tint and colour can materially impact aesthetics occupant mood and even clinical outcomes in health/care settings. Glass with good colour rendering offers similar colour tone inside as out (typically lighter/neutral tints and indicated by a high CRI value). SHGC considers total solar heat gain including both visible & non-visible light, modern spectrally selective & low'e' coatings allow the façade designer to optimise for heat gain and visible light transmission (i.e. glass can achieve SHGC=0.3 but still have VLT=45% and 'Good' CRI). Façade designer to consider CRI & VLT.
4. A slab on ground that does not have an in-slab heating or cooling system is considered to achieve a Total R-value of 2.0 as per NCC 2022 J4D7.

2.2 ADDITIONAL REQUIREMENTS

In addition to the construction commitments, there are further compliance requirement within Section J of the NCC 2022 Volume One that need to be achieved. These are typically part of architectural and building services specifications and drawings.

Part J4 – Building Fabric

- J4D3 (1-4) Thermal Construction – Material properties and general installation requirements for insulations.
- J4D3 (5)– Thermal break requirements.

Part J5 – Building Sealing

- J5D5 Windows and doors
- J5D6 Exhaust fans
- J5D7 Construction of roofs, walls and floors

3. COMPLIANCE SUMMARY

Per NCC 2022 Volume One Section J1V3 Part J1 compliance can be verified when the annual greenhouse gas emissions of the Proposed Building are not more than a Deemed-to-Satisfy (DTS) compliant Reference Building when calculated using compliant software and following the detailed procedure set out in the NCC.

The J1V3 process requires a digital representation of the project to be built and then its annual performance to be assessed on an hourly basis against a detailed building energy simulation weather file for a location appropriate to the site. This model considers the impacts of sun angle & solar intensity, air temperature & humidity and wind (amongst other aspects) for a representative year based on real and/or forecast weather from weather stations. For a more detailed explanation of the J1V3 modelling process please refer to Appendix F.

The image below provides a graphical representation of the project/building orientation sun path and shading effects for 1 of the 8760 hours in a representative year.

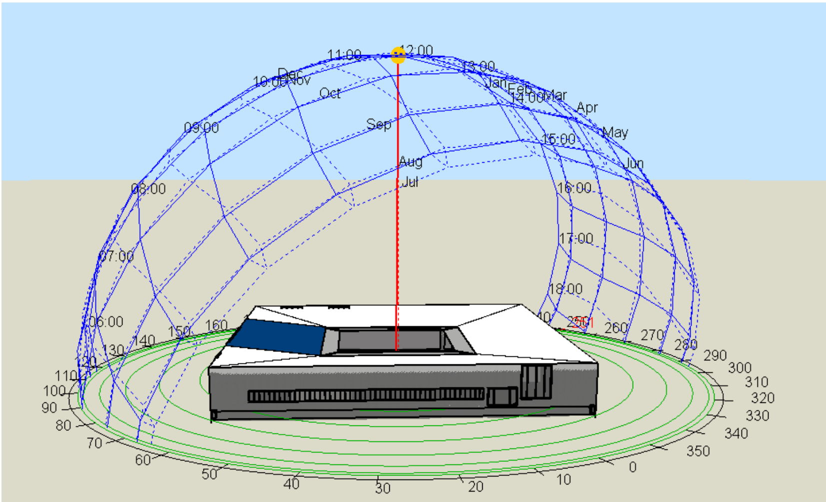

Figure 2 - Project/Building representation from the J1V3 Simulation Modelling Software

Compliance summary is tabulated below:

Table 2 – Compliance Summary

Elements	Reference Building	Proposed Building	Complies
Heating (Electricity) (kWh)	8,862.32	8,357.91	-
Cooling (Electricity) (kWh)	171,337.36	171,801.86	-
Room Electricity (kWh)	66,495.26	66,495.26	-
Lighting (kWh)	62,196.07	62,196.07	-
System Fans (kWh)	31,815.92	31,814.80	-
Renewable energy generated onsite	Nil	XXX	-
Electricity (kWh)	344,150.91	343,872.25	Yes
$CO_{2(e)}$ Emissions (kg)[1]	240,905.64	240,710.58	Yes
Thermal Comfort Performance Results (PMV)			COMPLIES

1. The listed carbon dioxide equivalent emissions (kg CO2(e)) figure is based on the average network carbon intensity figures. Please note that these figures are illustrative only based on historic grid information. The carbon intensity of the electricity supplied to this project will likely vary significantly over time depending on a variety of factors.

4. APPENDICES LIST

Appendix	Description	Status
A	Building Thermal Envelope Mark-Up	Included
B	Examples Of Construction Details	Included
C	DTS Calculations	Included
D	PMV Modelling Results	Included
E	Documentation Relied on For Assessment	Included
F	NCC 2022 J1V3 Modelling Details	Included
G	NCC 2022 J1V3 Operational Schedules	Included
H	NCC 2022 Parts J4 & J5 Excerpts	Included
I	NCC2022 Part J7D3A Number of Persons Accommodated	Included
J	NCC 2022 Table J6.2A Maximum Illumination Powe Density	Included

APPENDIX A – BUILDING THERMAL ENVELOPE MARK-UP

The attached markups indicate the proposed location of the building thermal envelope line adopted for the project JV3 assessment and must be read in conjunction with the table of building commitments. This envelope is intended to identify a location that will be practical for the installation of any necessary insulation and requires careful consideration by the building façade designers and mechanical HVAC system designers.

Ground Floor Plan
1 : 100

rev	date	description	initial
1	17/06/25	QS Issue #1	
2	08/07/2025	Background Issue	
3	05/08/2025	Background Issue	
4	11/08/2025	Coordination Issue	JB
5	29/08/2025	Coordination Issue	JB
6	08/09/2025	QS Issue	JB
7	19/09/2025	100% DD Issue	JB
8	13/10/2025	Revised Door Schedule	BG
9	27/10/2025	50% CD Issue	JB
10	21/11/2025	CD Coordination Issue	JB

JHA
MARKUP / SKETCH

DOCUMENT No.: Q250128 - ESD
DOCUMENT TITLE: BUILDING ENVELOPE MARKUP
Stanwell Office
DOCUMENT REV: B
DOCUMENT BY: R.H
DATE: 09/12/2025

WALL TOTAL R VALUE 1.9
GLAZING U:4.2 / SHGC:0.64
WALL TOTAL R VALUE 1.9
GLAZING U:4.7 / SHGC:0.59
ROOF TOTAL R VALUE: 5.9
FLOOR TOTAL R VALUE: 1.5

Drawings include colour information. A black & white copy may not have all the information. This background is coloured blue.

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client stanwell

project Stanwell New Projects Office
Switchyard Road,
Stanwell, QLD, 4702

title General Arrangement Plan

drawn bg
checked jb
date Nov 2025
scale 1 : 100 at A1
drawing 25010 / cd A-200001 / 10

1:100 at A1
0 1 2 3 4 5m

APPENDIX B – EXAMPLES OF CONSTRUCTION DETAILS

NB: the following examples are indicative based on the information included in the BCA NCC Section J Handbook / Guide. Proprietary details including thermal breaks in particular can have material impacts on final system performance and we recommend the final system performance be calculated and/or confirmed by the relevant manufacturers.

1. METAL ROOF

MINIMUM TOTAL R-VALUE: R5.9

RF-1

Layer	Material Description	R Value (m^2K/W)
1	Outside Air Film	0.04
2	Cladding (Solar Absorptance less than 0.45)	0
3	R4.3 Anticon Blanket 175mm + 150mm Ashgrid Spacer	3.94
4	Air Space	0.46
5	11kg/m3 - 75mm Acoustigard R1.7	1.70
6	Gypsum Plasterboard	0.06
7	Indoor Air Film	0.16
	Total System R-Value	6.3

RF-2

Layer	Material Description	R Value (m^2K/W)
1	Outside Air Film	0.04
2	Cladding (Solar Absorptance less than 0.45)	0
3	200 Z Purlin + 360UB + R2.5 insulation	0.74
4	Air Space	0.46
5	11kg/m3 - 75mm Acoustigard R1.7	1.70
6	Gypsum Plasterboard	0.06
7	Indoor Air Film	0.16
	Total System R-Value	3.1

Roof Type	%	R-value	Average
RF-1	86%	6.360	5.912
RF-2	14%	3.157	

2. LIGHTWEIGHT WALL

TYPICAL WALL – MINIMUM TOTAL R-VALUE: R1.9

Layer	Material Description	R Value (m^2K/W)
1	Outside Air Film	0.03
2	Cladding	0.00
3	6.0mm Cemintel Rigid Air Barrier	0.02
4	R4.0 140mm Gold Wall HP + R0.2 Thermal break	1.60
5	Unventilated airspace	0.13
6	Indoor Air Film (Vertical, Still)	0.12
	Total System R-Value	1.9

1. The specified thermal performance of the building fabric must be calculated including allowance for thermal bridging in accordance with AS/NZS 4589.2 as per NCC 2019 Amendment 1 Specification J1.5A requirements.

3. FLOOR

FLOOR – MINIMUM TOTAL R-VALUE: R1.5

Layer	Material Description	R Value (m^2K/W)
1	Outside Air Film	0.04
2	Cladding	0.00
3	31mm R1.3 Insulation under the joists	1.014
4	Airspace	0.22
5	18mm CFC Flooring	0.078
6	Indoor Air Film (Vertical, Still)	0.16
	Total System R-Value	1.5

APPENDIX C – DTS CALULATIONS

Please note that JHA uses its own developed spreadsheet, which is equivalent to the BCA NCC 2022 Volume One facade calculator. This tool helps in understanding and applying the NCC Volume One Part J4D6 Building Fabric Deemed-to-Satisfy (DTS) provisions.

Climate Zone	CZ 2
Class	Other
Display glazing?	No
Help note	1

Main Calculator

																Method 1		Method 2			Method 1	Method 2			QA
Override table						Façade Area										Solar Admittance				U-value			R-value		
			Window																						
Exposure	Description	Wall R-value	U-Value	SHGC	Total [m^s]	Total [m^2]	Wall [m^2]	Window [m^2]	Wall vs Wall Glazing Ratio	Total Extenral [m2]	Total Intenral [m2]	Wall Extenral [m2]	Wall Internal [m2]	Window Extenral [m2]	Window Internal [m2]	Max SA	Achieved SA	Max Er	Achieved Er	Max. U-Value	Achieved U-Value	Achieved U-Value	Min. R-Value	Achieved R-Value	Total wall length
N		1.40	4.0	0.45	282.6	282.6	125.8	156.8	45%	282.6	0.0	155.5	-29.7	127.1	29.7	0.13	0.149	71.64	82.09	2.0	2.54	1.97	1.0	1.40	78.5
E		1.40	4.0	0.45	162.0	162.0	92.7	69.3	57%	162.0	0.0	92.7	0.0	69.3	0.0	0.13	0.133	33.27	33.92	2.0	2.12		1.0	1.40	45
S		1.40	4.0	0.45	282.6	282.6	183.9	98.7	65%	282.6	0.0	213.6	-29.7	69.0	29.7	0.13	0.089	36.74	25.18	2.0	1.86		1.0	1.40	78.5
W		1.40	4.0	0.45	163.8	163.8	149.1	14.8	91%	163.8	0.0	149.1	0.0	14.8	0.0	0.13	0.031	0.00	0.00	2.0	1.01		1.4	1.40	45.5
																	SUM	141.65	141.19						

Glazing Input

ID	Description	Level	Exposure	Height [m]	Width [m]	_Int vs Ext	Area [m]	U-value	SHGC	P	H	P/H	G/H	Shading Multiplier	A*S*SHGC	A*U
1	W1	G	W	1.50	1.20	External	1.8			0.40	1.50	0.27	0.00	0.7467	0.60	0.00
2	W2	G	N	4.50	3.60	External	16.2			0.40	4.50	0.09	0.00	0.9111	6.64	0.00
3	W3	G	N	2.20	4.00	External	8.8			0.40	2.20	0.18	0.00	0.8182	3.24	0.00
4	W4	G	N	1.40	29.00	External	40.6			0.40	1.40	0.29	0.00	0.7314	13.36	0.00
5	W5	G	E	1.50	1.70	External	2.6			0.40	1.50	0.27	0.00	0.7467	0.86	0.00
6	W6	G	S	3.00	9.90	Internal	29.7					-	-	-	-	0.00
7	W7	G	S	3.00	20.50	External	61.5			2.00	3.60	0.56	0.17	0.8078	22.36	0.00
8	W8	G	E	3.00	20.50	External	61.5			2.00	3.60	0.56	0.17	0.6811	18.85	0.00
9	W9	G	N	3.00	20.50	External	61.5			2.00	3.60	0.56	0.17	0.6811	18.85	0.00
10	W10	G	N	3.00	9.90	Internal	29.7					-	-	-	-	0.00
11	W11	G	S	1.50	2.40	External	3.6			0.40	1.50	0.27	0.00	0.8367	1.36	0.00
12	W12	G	S	1.50	2.60	External	3.9			0.40	1.50	0.27	0.00	0.8367	1.47	0.00
13	W13	G	E	1.50	3.50	External	5.3			0.40	1.50	0.27	0.00	0.7467	1.76	0.00
14	W14	G	W	0.70	0.70	External	0.5			0.40	0.70	0.57	0.00	0.5271	0.12	0.00
15	W15	G	W	0.70	0.70	External	0.5			0.40	0.70	0.57	0.00	0.5271	0.12	0.00
16	W16	G	W	1.90	2.90	External	5.5			0.40	1.90	0.21	0.00	0.7916	1.96	0.00
17	W17	G	W	1.70	3.80	External	6.5			0.40	1.70	0.24	0.00	0.7718	2.24	0.00
18			Select				-					-	-	-	-	-

Wall Input

ID	Descript	Level	Exposure	_Int vs Ext	Height [m]	Width [m]	Area [m]
1	Wall 1	G	N	External	3.60	45.50	163.8
2	Wall 2	G	E	External	3.60	12.50	45.0
3	Wall 3	G	S	External	3.60	33.00	118.8
4	Wall 4	G	E	External	3.60	20.00	72.0
5	Wall 5	G	N	External	3.60	33.00	118.8
6	Wall 6	G	E	External	3.60	12.50	45.0
7	Wall 7	G	S	External	3.60	45.50	163.8
8	Wall 8	G	W	External	3.60	45.50	163.8
9			Select				-
10			Select				-
11			Select				-
12			Select				-
13			Select				-
14			Select				-
15			Select				-
16			Select				-
17			Select				-
18			Select				-

APPENDIX D – PMV MODELLING RESULTS

The results show 100% of floor areas achieve a thermal comfort level of between a Predicted Mean Vote (PMV) of -1 to +1 for not less than 98% of the annual hours of operation of the building. Therefore, PMV modelling results demonstrate that the proposed building meets the J1V3 Verification Method thermal comfort level requirements.

Table 1 – J1V3 Thermal Comfort Performance Results per zone

Locations	Area (m2)	PMV (% hours in range)			Meets J1V3 (1)(b) criteria	Compliant Areas (m2)
		< -1.0	≥-1.0 & ≤1.0	> 1.0		
First Aid	12.7	0.0%	100.0%	0.0%	Y	12.7
Parents Room	9.6	0.0%	100.0%	0.0%	Y	9.6
Dining	155.6	0.0%	100.0%	0.0%	Y	155.6
Boardroom	73.8	0.0%	100.0%	0.0%	Y	73.8
Training	73.5	0.0%	100.0%	0.0%	Y	73.5
Training Kitchen	65.9	0.0%	100.0%	0.0%	Y	65.9
Meeting 6	13	0.0%	100.0%	0.0%	Y	13
Meeting 5	13	0.0%	100.0%	0.0%	Y	13
Meeting 2	7.6	0.0%	100.0%	0.0%	Y	7.6
Meeting 4	12.8	0.0%	100.0%	0.0%	Y	12.8
Office	7.6	0.0%	100.0%	0.0%	Y	7.6
Workspace	738.3	0.0%	100.0%	0.0%	Y	738.3
Meeting 1	9	0.0%	100.0%	0.0%	Y	9
Meeting 3	20.8	0.0%	100.0%	0.0%	Y	20.8
Total	1,213.2					100%

	PMV range	% Nominated floor area with min 98% of hours in PMV range	Compliant
Proposed Thermal Comfort	-1 to 1	100%	Yes

APPENDIX E – DOCUMENTATION RELIED ON FOR ASSESSMENT

The following documentation was relied on in the preparation of the report. JHA note that the design documentation is typically amended and developed ongoingly after finalisation of a NCC Section J Compliance Assessment to incorporate additional construction and/or proprietary material details. This is normal/expected and the Section J Assessment remains valid provided that the designers can confirm that the minimum requirements of the assessment are still achieved.

Drawing	Brief Description	Issue Date
25010 / cd A-200001 / 10	*General Arrangement Plan*	*Nov 2025*
25010 / cd A-200002 / 8	*Roof Plan*	*Nov 2025*
25010 / cd A-200003 / 8	*Roof Cavity Plan*	*Nov 2025*
25010 / cd A-260001 / 8	*General Elevations*	*Nov 2025*
25010 / cd A-260002 / 8	*General Elevations*	*Nov 2025*
25010 / cd A-270001 / 8	*General Sections*	*Nov 2025*
25010 / cd A-270002 / 6	*General Sections*	*Nov 2025*
	Revit Model	

APPENDIX F – NCC 2022 J1V3 MODELLING DETAILS

GENERAL

As per the NCC 2022 Section J1V3, Part J1 compliance can be verified when the annual greenhouse gas emissions of the Proposed Building are not more than a Deemed-to-Satisfy (DTS) compliant Reference Building when:

- The proposed building is modelled with the proposed services;
- The proposed building is modelled with the same services as the reference building;
- In the proposed building, a thermal comfort level between a Predicted Mean Vote of -1 to +1 is achieved across not less than 95% of the floor area of all occupied zones for not less than 98% of the annual hours of operation of the building;
- The annual greenhouse gas emissions of the proposed buildings may be offset by renewable energy generated and used on site.
- The building complies with additional requirements in specification 33 including:
    - J4D3 General Thermal Construction
    - J4D7 (2) and J4D7 (3) Floor Edge Insulation

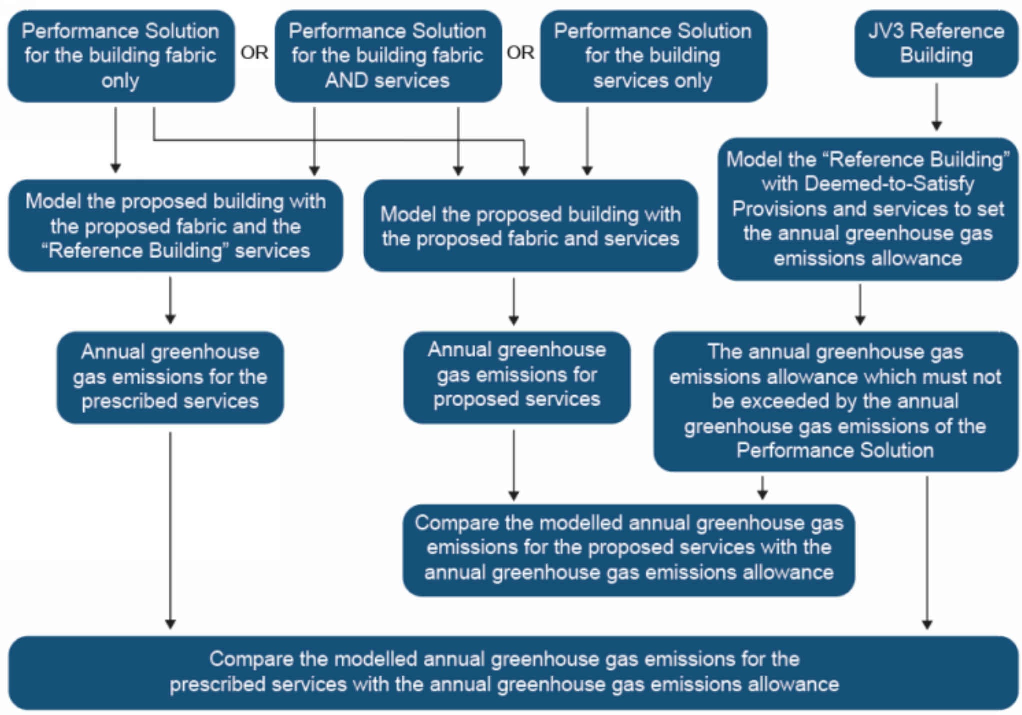

Figure 11 – BCA Verification Method of J1V3

MODELLING ASSUMPTIONS

In accordance with the NCC Section JVb 4(h), modelling of hot water energy consumption and lifts have been excluded.

In accordance with the NCC Section JVc, the following modelling profiles have been incorporated into the J1V3 modelling.

INTERNAL LOADS

The following occupancy, artificial lighting and equipment load have been incorporated into the J1V3 modelling as per NCC 2022 Specification 35 for Part J1V3.

Table 2 – Internal Loads

APPLICATION	INTERNAL HEAT GAIN
Occupancy Heat Gain	Sensible Heat: 75 W Latent Heat: 55W Occupancy Density: As determined by NCC 2022 Volume 1 Part D2D18 (Refer to Appendix J)
Artificial Lighting	As determined by NCC 2022 Part J7 (Refer to Appendix K)
Appliances and Equipment	Sensible Heat: 11 W/m^2

REFERENCE SERVICES

The following table indicated the modelling parameters used in the J1V3 modelling in accordance with NCC 2022 Specification 34 for Part J1V3.

Table 3 – Reference Services

APPLICATION	PARAMETERS
Air Conditioning	Cooling Setpoint: 24 ^{o}CDB Heating Setpoint: 21 ^{o}CDB Cooling / Heating COP: 3.1 (as per MEPS)
Mechanical Ventilation Rate	As per NCC Part F4
Infiltration	0.7 ACH for spaces without mechanical supply ventilation 0.35 ACH for all other times

APPENDIX G - NCC 2022 JV3 OPERATIONAL SCHEDULES

The following modelling profile have been incorporated into the J1V3 modelling as per NCC 2022 Specification 35 for Part J1V3.

Table 4 – Operation Profiles for Class 5 Office building – Weekdays

Time Period	Occupancy	Artificial Lighting	Appliances and Equipment	Air Conditioning
12:00am to 1:00am	0%	15%	25%	Off
1:00am to 2:00am	0%	15%	25%	Off
2:00am to 3:00am	0%	15%	25%	Off
3:00am to 4:00am	0%	15%	25%	Off
4:00am to 5:00am	0%	15%	25%	Off
5:00am to 6:00am	0%	15%	25%	Off
6:00am to 7:00am	0%	15%	25%	Off
7:00am to 8:00am	10%	40%	65%	On
8:00am to 9:00am	20%	90%	80%	On
9:00am to 10:00am	70%	100%	100%	On
10:00am to 11:00am	70%	100%	100%	On
11:00am to 12:00pm	70%	100%	100%	On
12:00pm to 1:00pm	70%	100%	100%	On
1:00pm to 2:00pm	70%	100%	100%	On
2:00pm to 3:00pm	70%	100%	100%	On
3:00pm to 4:00pm	70%	100%	100%	On
4:00pm to 5:00pm	70%	100%	100%	On
5:00pm to 6:00pm	35%	80%	80%	On
6:00pm to 7:00pm	10%	60%	65%	Off
7:00pm to 8:00pm	5%	60%	55%	Off
8:00pm to 9:00pm	5%	50%	25%	Off
9:00pm to 10:00pm	0%	15%	25%	Off
10:00pm to 11:00pm	0%	15%	25%	Off
11:00pm to 12:00am	0%	15%	25%	Off

Table 5– Operation Profiles for Class 5 Office building - Weekends

Time Period	Occupancy	Artificial Lighting	Appliances and Equipment	Air Conditioning
12:00am to 1:00am	0%	15%	25%	Off
1:00am to 2:00am	0%	15%	25%	Off
2:00am to 3:00am	0%	15%	25%	Off
3:00am to 4:00am	0%	15%	25%	Off
4:00am to 5:00am	0%	15%	25%	Off
5:00am to 6:00am	0%	15%	25%	Off
6:00am to 7:00am	0%	15%	25%	Off
7:00am to 8:00am	0%	15%	25%	Off
8:00am to 9:00am	5%	25%	25%	Off
9:00am to 10:00am	5%	25%	25%	Off
10:00am to 11:00am	5%	25%	25%	Off
11:00am to 12:00pm	5%	25%	25%	Off
12:00pm to 1:00pm	5%	25%	25%	Off
1:00pm to 2:00pm	5%	25%	25%	Off
2:00pm to 3:00pm	5%	25%	25%	Off
3:00pm to 4:00pm	5%	25%	25%	Off
4:00pm to 5:00pm	5%	25%	25%	Off
5:00pm to 6:00pm	0%	15%	25%	Off
6:00pm to 7:00pm	0%	15%	25%	Off
7:00pm to 8:00pm	0%	15%	25%	Off
8:00pm to 9:00pm	0%	15%	25%	Off
9:00pm to 10:00pm	0%	15%	25%	Off
10:00pm to 11:00pm	0%	15%	25%	Off
11:00pm to 12:00am	0%	15%	25%	Off

APPENDIX H – NCC 2022 PARTS J4 & J5 EXCERPTS

Part J4 Building fabric

NT Part J4

Introduction to this Part

This Part contains *Deemed-to-Satisfy Provisions* for compliance with Part J1. It sets out provisions for the building *envelope* including roofs, ceilings, roof lights, walls, *glazing* and floors.

Notes

From 1 May 2023 to 30 September 2023 Section J of NCC 2019 Volume One Amendment 1 may apply instead of Section J of NCC 2022 Volume One. From 1 October 2023 Section J of NCC 2022 Volume One applies.

Notes: New South Wales Section J Energy Efficiency

(1) For a Class 2 building or a Class 4 part of a building, where a relevant *development consent* or an application for a complying development certificate requires compliance with a BASIX Single Dwelling or Multi Dwelling Certificate issued under Version 3.0 or earlier, NSW Section J of NCC 2019 Volume One Amendment 1 applies.

(2) For a Class 2 building or a Class 4 part of a building, where a relevant *development consent* or an application for a complying development certificate requires compliance with a BASIX Single Dwelling or Multi Dwelling Certificate issued under Version 4.0 or later, Section J of NCC 2022 Volume One applies.

(3) For a Class 2 building or a Class 4 part of a building, where a relevant *development consent* or an application for a complying development certificate requires compliance with a BASIX Alterations and Additions Certificate, NSW Section J of NCC 2019 Volume One Amendment 1 applies.

(4) For a Class 3 building or Class 5 to 9 building:

- (i) From 1 May 2023 to 30 September 2023 NSW Section J of NCC 2019 Volume One Amendment 1 may apply instead of Section J of NCC 2022 Volume One.
- (ii) From 1 October 2023 Section J of NCC 2022 Volume One applies.

Notes: Tasmania Section J Energy Efficiency

In Tasmania, for a Class 2 building and Class 4 part of a building, Section J is replaced with Section J of BCA 2019 Amendment 1.

Deemed-to-Satisfy Provisions

J4D1 Deemed-to-Satisfy Provisions

[2019: J1.0]

NSW J4D1(1)

(1) Where a *Deemed-to-Satisfy Solution* is proposed, *Performance Requirements* J1P1 to J1P4 are satisfied by complying with—

- (a) J2D2; and
- (b) J3D2 to J3D15; and
- (c) J4D2 to J4D7; and
- (d) J5D2 to J5D8; and
- (e) J6D2 to J6D13; and
- (f) J7D2 to J7D9; and
- (g) J8D2 to J8D4; and

(h) J9D2 to J9D5.

(2) Where a *Performance Solution* is proposed, the relevant *Performance Requirements* must be determined in accordance with A2G2(3) and A2G4(3) as applicable.

NSW J4D2

J4D2 Application of Part

[2019: J1.1]

The *Deemed-to-Satisfy Provisions* of this Part apply to building elements forming the *envelope* of a Class 2 to 9 building other than J4D3(5), J4D4, J4D5, J4D6 and J4D7 which do not apply to a Class 2 *sole-occupancy unit* or a Class 4 part of a building.

NSW J4D3

J4D3 Thermal construction — general

[2019: J1.2]

(1) Where *required*, insulation must comply with AS/NZS 4859.1 and be installed so that it—
   (a) abuts or overlaps adjoining insulation other than at supporting members such as studs, noggings, joists, furring channels and the like where the insulation must be against the member; and
   (b) forms a continuous barrier with ceilings, walls, bulkheads, floors or the like that inherently contribute to the thermal barrier; and
   (c) does not affect the safe or effective operation of a *service* or fitting.

(2) Where *required*, *reflective insulation* must be installed with—
   (a) the necessary airspace to achieve the *required R-Value* between a reflective side of the *reflective insulation* and a building lining or cladding; and
   (b) the *reflective insulation* closely fitted against any penetration, door or *window* opening; and
   (c) the *reflective insulation* adequately supported by framing members; and
   (d) each adjoining sheet of roll membrane being—
      (i) overlapped not less than 50 mm; or
      (ii) taped together.

(3) Where *required*, bulk insulation must be installed so that—
   (a) it maintains its position and thickness, other than where it is compressed between cladding and supporting members, water pipes, electrical cabling or the like; and
   (b) in a ceiling, where there is no bulk insulation or *reflective insulation* in the wall beneath, it overlaps the wall by not less than 50 mm.

(4) Roof, ceiling, wall and floor materials, and associated surfaces are deemed to have the thermal properties listed in Specification 36.

(5) The *required Total R-Value* and *Total System U-Value*, including allowance for thermal bridging, must be—
   (a) calculated in accordance with AS/NZS 4859.2 for a roof or floor; or
   (b) determined in accordance with Specification 37 for *wall-glazing construction*; or
   (c) determined in accordance with Specification 39 or Section 3.5 of CIBSE Guide A for soil or sub-floor spaces.

J4D4 Roof and ceiling construction

[2019: J1.3]

(1) A roof or ceiling must achieve a *Total R-Value* greater than or equal to—
   (a) in *climate zones* 1, 2, 3, 4 and 5, R3.7 for a downward direction of heat flow; and

(b) in *climate zone* 6, R3.2 for a downward direction of heat flow; and
(c) in *climate zone* 7, R3.7 for an upward direction of heat flow; and
(d) in *climate zone* 8, R4.8 for an upward direction of heat flow.

(2) In *climate zones* 1, 2, 3, 4, 5, 6 and 7, the solar absorptance of the upper surface of a roof must be not more than 0.45.

J4D5 Roof lights

[2019: J1.4]

Roof lights must have—

(a) a total area of not more than 5% of the *floor area* of the room or space served; and
(b) transparent and translucent elements, including any imperforate ceiling diffuser, with a combined performance of—
  (i) for *Total system SHGC*, in accordance with Table J4D5; and
  (ii) for *Total system U-Value*, not more than U3.9.

Table J4D5: Roof lights – Total system SHGC

Roof light shaft index Note 1	Total area of *roof lights* up to 3.5% of the *floor area* of the room or space	Total area of *roof lights* more than 3.5% and up to 5% of the *floor area* of the room or space
<1.0	≤ 0.45	≤ 0.29
≥ 1.0 to < 2.5	≤ 0.51	≤ 0.33
≥ 2.5	≤ 0.76	≤ 0.49

Table Notes

(1) The *roof light* shaft index is determined by measuring the distance from the centre of the shaft at the roof to the centre of the shaft at the ceiling level and dividing it by the average internal dimension of the shaft opening at the ceiling level (or the diameter for a circular shaft) in the same units of measurement.
(2) The area of a *roof light* is the area of the roof opening that allows light to enter the building.
(3) The total area of *roof lights* is the combined area for all *roof lights* serving the room or space.

NSW J4D6

J4D6 Walls and glazing

[2019: J1.5]

(1) The *Total System U-Value* of *wall-glazing construction*, including *wall-glazing construction* which wholly or partly forms the *envelope* internally, must not be greater than—
  (a) for a Class 2 common area, a Class 5, 6, 7, 8 or 9b building or a Class 9a building other than a *ward area*, U2.0; and
  (b) for a Class 3 or 9c building or a Class 9a *ward area*—
    (i) in *climate zones* 1, 3, 4, 6 or 7, U1.1; or
    (ii) in *climate zones* 2 or 5, U2.0; or
    (iii) in *climate zone* 8, U0.9.

(2) The *Total System U-Value* of *display glazing* must not be greater than U5.8.
(3) The *Total System U-Value* of *wall-glazing construction* must be calculated in accordance with Specification 37.
(4) Wall components of a *wall-glazing construction* must achieve a minimum *Total R-Value* of—
  (a) where the wall is less than 80% of the area of the *wall-glazing construction*, R1.0; or

(b) where the wall is 80% or more of the area of the *wall-glazing construction*, the value specified in Table J4D6a.

(5) The *solar admittance* of externally facing *wall-glazing construction*, excluding *wall-glazing construction* which is wholly internal, must not be greater than—

(a) for a Class 2 common area, a Class 5, 6, 7, 8 or 9b building or a Class 9a building other than a *ward area*, the values specified in Table J4D6b; and

(b) for a Class 3 or 9c building or a Class 9a *ward area*, the values specified in Table J4D6c.

(6) The *solar admittance* of a *wall-glazing construction* must be calculated in accordance with Specification 37.

(7) The *Total system SHGC* of *display glazing* must not be greater than 0.81 divided by the applicable shading factor specified in S37C7.

Table J4D6a: Minimum wall Total R-Value - Wall area 80% or more of wall-glazing construction area

Climate zone	Class 2 common area, Class 5, 6, 7, 8 or 9b building or a Class 9a building other than a *ward area*	Class 3 or 9c building or Class 9a *ward area*
1	2.4	3.3
2	1.4	1.4
3	1.4	3.3
4	1.4	2.8
5	1.4	1.4
6	1.4	2.8
7	1.4	2.8
8	1.4	3.8

Table J4D6b: Maximum wall-glazing construction solar admittance - Class 2 common area, Class 5, 6, 7, 8 or 9b building or Class 9a building other than a ward area

Climate zone	Eastern aspect *solar admittance*	Northern aspect *solar admittance*	Southern aspect *solar admittance*	Western aspect *solar admittance*
1	0.12	0.12	0.12	0.12
2	0.13	0.13	0.13	0.13
3	0.16	0.16	0.16	0.16
4	0.13	0.13	0.13	0.13
5	0.13	0.13	0.13	0.13
6	0.13	0.13	0.13	0.13
7	0.13	0.13	0.13	0.13
8	0.2	0.2	0.42	0.36

Table J4D6c: Maximum wall-glazing construction solar admittance - Class 3 or 9c building or Class 9a ward area

Climate zone	Eastern aspect *solar admittance*	Northern aspect *solar admittance*	Southern aspect *solar admittance*	Western aspect *solar admittance*
1	0.07	0.07	0.10	0.07
2	0.10	0.10	0.10	0.10
3	0.07	0.07	0.07	0.07
4	0.07	0.07	0.07	0.07
5	0.10	0.10	0.10	0.10
6	0.07	0.07	0.07	0.07

Climate zone	Eastern aspect *solar admittance*	Northern aspect *solar admittance*	Southern aspect *solar admittance*	Western aspect *solar admittance*
7	0.07	0.07	0.08	0.07
8	0.08	0.08	0.08	0.08

J4D7 Floors

[2019: J1.6]

(1) A floor must achieve the *Total R-Value* specified in Table J4D7.

(2) For the purposes of (1), a slab-on-ground that does not have an in-slab heating or cooling system is considered to achieve a *Total R-Value* of R2.0, except—

(a) in *climate zone* 8; or

(b) a Class 3, Class 9a *ward area* or Class 9b building in *climate zone* 7 that has a *floor area* to floor perimeter ratio of less than or equal to 2.

(3) A floor must be insulated around the vertical edge of its perimeter with insulation having an *R-Value* greater than or equal to 1.0 when the floor—

(a) is a concrete slab-on-ground in *climate zone* 8; or

(b) has an in-slab or in-screed heating or cooling system, except where used solely in a bathroom, amenity area or the like.

(4) Insulation *required* by (3) for a concrete slab-on-ground must—

(a) be *water resistant*; and

(b) be continuous from the adjacent finished ground level—

(i) to a depth not less than 300 mm; or

(ii) for the full depth of the vertical edge of the concrete slab-on-ground.

Table J4D7: Floors – Minimum Total R-Value

Location	*Climate zone* 1— upwards heat flow	*Climate zones* 2 and 3 — upwards and downwards heat flow	*Climate zones* 4, 5, 6 and 7 — downwards heat flow	*Climate zone* 8 — downwards heat flow
A floor without an in-slab heating or cooling system	2.0	2.0	2.0	3.5
A floor with an in-slab heating or cooling system	3.25	3.25	3.25	4.75

Table Notes

For the purpose of calculating the *Total R-Value* of a floor, the sub-floor and soil *R-Value* must be calculated in accordance with Specification 39 or Section 3.5 of CIBSE Guide A.

Part J5 Building sealing

NT Part J5

Introduction to this Part

This Part contains *Deemed-to-Satisfy Provisions* for compliance with Part J1. It sets out provisions for the sealing of a building's *glazing*, doors, exhaust fans and the like in order to increase thermal comfort for occupants and reduce the energy consumption of any installed *air-conditioning* systems.

Notes

From 1 May 2023 to 30 September 2023 Section J of NCC 2019 Volume One Amendment 1 may apply instead of Section J of NCC 2022 Volume One. From 1 October 2023 Section J of NCC 2022 Volume One applies.

Notes: New South Wales Section J Energy Efficiency

(1) For a Class 2 building or a Class 4 part of a building, where a relevant *development consent* or an application for a complying development certificate requires compliance with a BASIX Single Dwelling or Multi Dwelling Certificate issued under Version 3.0 or earlier, NSW Section J of NCC 2019 Volume One Amendment 1 applies.

(2) For a Class 2 building or a Class 4 part of a building, where a relevant *development consent* or an application for a complying development certificate requires compliance with a BASIX Single Dwelling or Multi Dwelling Certificate issued under Version 4.0 or later, Section J of NCC 2022 Volume One applies.

(3) For a Class 2 building or a Class 4 part of a building, where a relevant *development consent* or an application for a complying development certificate requires compliance with a BASIX Alterations and Additions Certificate, NSW Section J of NCC 2019 Volume One Amendment 1 applies.

(4) For a Class 3 building or Class 5 to 9 building:

  (i) From 1 May 2023 to 30 September 2023 NSW Section J of NCC 2019 Volume One Amendment 1 may apply instead of Section J of NCC 2022 Volume One.

  (ii) From 1 October 2023 Section J of NCC 2022 Volume One applies.

Notes: Tasmania Section J Energy Efficiency

In Tasmania, for a Class 2 building and Class 4 part of a building, Section J is replaced with Section J of BCA 2019 Amendment 1.

Deemed-to-Satisfy Provisions

J5D1 Deemed-to-Satisfy Provisions

[2019: J3.0]

NSW J5D1(1)

(1) Where a *Deemed-to-Satisfy Solution* is proposed, *Performance Requirements* J1P1 to J1P4 are satisfied by complying with—

  (a) J2D2; and

  (b) J3D2 to J3D15; and

  (c) J4D2 to J4D7; and

  (d) J5D2 to J5D8; and

  (e) J6D2 to J6D13; and

  (f) J7D2 to J7D9; and

(g) J8D2 to J8D4; and

(h) J9D2 to J9D5.

(2) Where a *Performance Solution* is proposed, the relevant *Performance Requirements* must be determined in accordance with A2G2(3) and A2G4(3) as applicable.

NSW J5D2

J5D2 Application of Part

[2019: J3.1]

The *Deemed-to-Satisfy Provisions* of this Part apply to elements forming the *envelope* of a Class 2 to 9 building, other than—

(a) a building in *climate zones* 1, 2, 3 and 5 where the only means of *air-conditioning* is by using an evaporative cooler; or

(b) a permanent building opening, in a space where a gas appliance is located, that is necessary for the safe operation of a gas appliance; or

(c) a building or space where the mechanical ventilation *required* by Part F6 provides sufficient pressurisation to prevent infiltration.

J5D3 Chimneys and flues

[2019: J3.2]

The chimney or flue of an open solid-fuel burning appliance must be provided with a damper or flap that can be closed to seal the chimney or flue.

J5D4 Roof lights

[2019: J3.3]

(1) A *roof light* must be sealed, or capable of being sealed, when serving—

(a) a *conditioned space*; or

(b) a *habitable room* in *climate zones* 4, 5, 6, 7 or 8.

(2) A *roof light required* by (1) to be sealed, or capable of being sealed, must be constructed with—

(a) an imperforate ceiling diffuser or the like installed at the ceiling or internal lining level; or

(b) a weatherproof seal; or

(c) a shutter system readily operated either manually, mechanically or electronically by the occupant.

NSW J5D5

J5D5 Windows and doors

[2019: J3.4]

(1) A door, openable *window* or the like must be sealed—

(a) when forming part of the *envelope*; or

(b) in *climate zones* 4, 5, 6, 7 or 8.

(2) The requirements of (1) do not apply to—

(a) a *window* complying with AS 2047; or

(b) a fire door or smoke door; or

(c) a roller shutter door, roller shutter grille or other security door or device installed only for out-of-hours security.

(3) A seal to restrict air infiltration—
   (a) for the bottom edge of a door, must be a draft protection device; and
   (b) for the other edges of a door or the edges of an openable *window* or other such opening, may be a foam or rubber compression strip, fibrous seal or the like.
(4) An entrance to a building, if leading to a *conditioned space* must have an airlock, *self-closing* door, *rapid roller door*, revolving door or the like, other than—
   (a) where the *conditioned space* has a *floor area* of not more than 50 m^2; or
   (b) where a café, restaurant, open front shop or the like has—
      (i) a 3 m deep un-conditioned zone between the main entrance, including an open front, and the *conditioned space*; and
      (ii) at all other entrances to the café, restaurant, open front shop or the like, *self-closing* doors.
(5) A loading dock entrance, if leading to a *conditioned space*, must be fitted with a *rapid roller door* or the like.

J5D6 Exhaust fans

[2019: J3.5]

An exhaust fan must be fitted with a sealing device such as a self-closing damper or the like when serving—
   (a) a *conditioned space*; or
   (b) a *habitable room* in *climate zones* 4, 5, 6, 7 or 8.

J5D7 Construction of ceilings, walls and floors

[2019: J3.6]

(1) Ceilings, walls, floors and any opening such as a *window* frame, door frame, *roof light* frame or the like must be constructed to minimise air leakage in accordance with (2)—
   (a) when forming part of the *envelope*; or
   (b) in *climate zones* 4, 5, 6, 7 or 8.
(2) Construction *required* by (1) must be—
   (a) enclosed by internal lining systems that are close fitting at ceiling, wall and floor junctions; or
   (b) sealed at junctions and penetrations with—
      (i) close fitting architrave, skirting or cornice; or
      (ii) expanding foam, rubber compressible strip, caulking or the like.
(3) The requirements of (1) do not apply to openings, grilles or the like *required* for smoke hazard management.

J5D8 Evaporative coolers

[2019: J3.7]

An evaporative cooler must be fitted with a self-closing damper or the like—
   (a) when serving a heated space; or
   (b) in *climate zones* 4, 5, 6, 7 or 8.

APPENDIX I – NCC2022 VOLUME 1 PART D2D18

Type of Use	Area per person, m^2
Art gallery, exhibition area, museum	4
Bar – standing, to 1.5m from edge of bar top	0.5
Bar – other	1
Board Room	2
Boarding house	15
Cafe, church, dining room	1
Carpark	30
Computer room	25
Court room – judicial area	10
Court room – public seating	1
Dance floor	0.5
Dormitory	5
Early childhood centre	4
Factory – machine shop, fitting shop, or like place for cutting grading, finishing or fitting of metals or glass	5
Factory – fabrication of structural steelwork, manufacture of vehicles or bulky products	50
Gymnasium	3
Hostel, hotel, motel, guest house	15
Indoor sports stadium – arena	10
Kiosk	1
Kitchen, laboratory, laundry	10
Library – reading space	2
Library – storage space	30
Office – typewriting, document copying	10
Patient care areas	10
Plant room – ventilation, electrical or other service units	30
Plant room – boilers or power plant	50
Reading room	2
Restaurant	1
School – general classroom	2
School – multi-purpose hall	1
School – staff room	10
School – trade and practical area – primary	4
School – trade and practical area – secondary	As for workshop
Shop – space for sale of goods – entry direct from open air or lower level	3
Shop – space for sale of goods – all other levels	5
Showroom – display area, covered mall or arcade	5
Skating rink, based on rink area	1.5
Spectator stand, audience viewing area – standing	0.3
Spectator stand, audience viewing area – removable seating	1
Spectator stand, audience viewing area – fixed seating	Number of seats
Spectator stand, audience viewing area – bench seating	450mm/person
Storage space	30
Swimming pool, based on pool area	1.5
Switch room, transformer room	30
Telephone exchange – private	30
Theatre and public hall	1
Theatre dressing room	4
Transport terminal	2
Workshop – for maintenance staff	30
Workshop – for manufacturing processes	As for factory

APPENDIX J – NCC 2022 TABLE J7D3A

Space	Maximum illumination power density (W/m 2)
Auditorium, church and public hall	8
Board room and conference room	5
Carpark - general	2
Carpark - entry zone (first 15 m of travel) during the daytime	11.5
Carpark - entry zone (next 4 m of travel) during the day	2.5
Carpark - entry zone (first 20 m of travel) during nighttime	2.5
Common rooms, spaces and corridors in a Class 2 building	4.5
Control room, switch room and the like - intermittent monitoring	3
Control room, switch room and the like - constant monitoring	4.5
Corridors	5
Courtroom	4.5
Dormitory of a Class 3 building used for sleeping only	3
Dormitory of a Class 3 building used for sleeping and study	4
Entry lobby from outside the building	9
Health-care - infants' and children's wards and emergency department	4
Health-care - examination room	4.5
Health-care - examination room in intensive care and high dependency ward	6
Health-care - all other patient care areas including wards and corridors	2.5
Kitchen and food preparation area	4
Laboratory - artificially lit to an ambient level of 400 lx or more	6
Library - stack and shelving area	2.5
Library - reading room and general areas	4.5
Lounge area for communal use in a Class 3 or 9c building	4.5
Museum and gallery - circulation, cleaning and service lighting	2.5
Office - artificially lit to an ambient level of 200 lx or more	4.5
Office - artificially lit to an ambient level of less than 200 lx	2.5
Plant room where an average of 160 lx vertical illuminance is required on a vertical panel such as in switch rooms	4
Plant rooms with a horizontal illuminance target of 80 lx	2
Restaurant, café, bar, hotel lounge and a space for the serving and consumption of food or drinks	14
Retail space including a museum and gallery whose purpose is the sale of objects	14
School - general purpose learning areas and tutorial rooms	4.5
Sole-occupancy unit of a Class 3 or 9c building	5
Storage	1.5
Service area, cleaner's room and the like	1.5
Toilet, locker room, staff room, rest room and the like	3
Wholesale storage area with a vertical illuminance target of 160 lx	4
Stairways, including fire-isolated stairways	2
Lift cars	3